Steganography Tool for Data Hiding

Introduction

Steganography is the technique of hiding secret data inside another file such as an image. This project demonstrates secure communication using hidden messages.

Abstract

This project is a Python-based steganography tool that hides text inside an image using Least Significant Bit (LSB) technique. The hidden message can only be retrieved using the decoding process, ensuring privacy and secrecy.

Tools Used

- Python
- PIL (Pillow) library

Steps Involved

1. Load an image file.
2. Convert secret message into binary format.
3. Embed the message into image pixels (LSB method).
4. Save the modified image.
5. Decode the image to extract hidden message.

Conclusion

This project highlights how data can be hidden securely within images. It demonstrates an interesting method of protecting information beyond traditional encryption.